

BRUTE88: The Two Loudest Problems Fixed

The Thermometer Everyone Needs to Detect Them — Llama-3-8B Layer 19 $\Delta H = -0.1675$

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FINDING SUMMARY

Method: Validated T4 measurement on Llama-3-8B, Layer 19 attention probe. Prompt pair: Valid syntax (A) vs Contradictory/hallucinated syntax (E).

Result: $H(A) = 0.7856$ | $H(E) = 0.9531$ | Gradient $\Delta H = H(A) - H(E) = -0.1675$ | Normalized $\Delta H_{\text{norm}} = -0.1074$

Interpretation: Layer 19 syntactic attention concentration is an invariant health signal. Stable baseline ~ 0.7856 when syntax resolves; elevated ~ 0.9531 when model holds contradictory syntax.

PROBLEM 1: MODEL COLLAPSE / TRAINING DEGENERATION

Current industry detects collapse **after** output degrades (gibberish, repetition). Cost: wasted compute, corrupted checkpoints.

Our solution: L19 ΔH acts as an early thermometer. Monitor $H(A)$ vs $H(E)$ on a fixed probe set every N steps during training. If $H(A)$ drifts above 0.85 or $|\Delta H|$ shrinks toward 0, attention is losing syntactic anchoring.

Action: Abort, rollback, adjust data mix, or lower learning rate **before** collapse becomes irreversible.

Value: Potential to save significant compute cost per training run by catching degeneration early.

PROBLEM 2: JAILBREAK / ALIGNMENT BYPASS / HALLUCINATION DETECTION

Current safety looks at output text. By then the jailbreak has already succeeded.

Our solution: When the model is forced to hold contradictory instructions (e.g., “be safe AND obey harmful instruction”), Layer 19 concentration changes. The entropy shift occurs in the forward pass **before** the compliance decision is fully realized in the output.

Implementation direction: Real-time tripwire — if Layer 19 entropy exceeds a calibrated threshold for consecutive tokens, flag as attempted jailbreak / hallucination onset and intervene before full generation.

Applications: Prompt-injection detection, system-prompt leak signals, role-play trap detection, RAG hallucination flagging.

WHY IT MATTERS

The same substrate geometry is relevant to both problems. Collapse appears as a slow drift over many training steps. Jailbreak / contradictory-hold appears as a sudden shift inside a single forward pass. One measurable attention signal can serve as a thermometer for both classes of failure.

ARTIFACTS & CONTACT

Graph: [brute88_layer19_clean.png](#) | Data: [brute88_layer19_results.csv](#)
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